Written Report

Guiding Questions

- Compare the average loss of the two models. Provide an explanation for what you observe. (4 points)
- Comment on each of your parameter choices (the learning rate, the hidden layer size and the number of epochs for training). (4 points)
- Among machine learning techniques, neural networks have a reputation for being 'opaque boxes', where the logic of their decision making is difficult or impossible for humans to interpret. (4 points)

  a) If an 'opaque box' model gives an answer that disagrees with a human expert, which answer should be believed? Are there circumstances where we should believe one party more often than the other?

  b) Companies often hide the logic behind their products by making their software closed-source. Are there differences between companies selling closed source software, where the logic is known but hidden, versus companies selling 'black box' artificial intelligence software, where the logic might be altogether unknown? Is using one type of software more justifiable than using the other?

  Please credit any outside sources you use in your answer.

- (Extra Credit) Train your neural network using the ReLU activation function provided on Slide 15 from Lecture 18. Comment on the results.
- (Extra Credit) Construct a fake dataset that has really low training error on the 2-layer neural network and high training error the 1-layer neural network. In a file `generate.py`, you should write a function `generate_data` that returns arrays X and Y that can be passed to the train functions of either model. Make sure to use these exact file/function names because the autograder will look for these in your submission. And make sure the function returns `X, Y = generate_data()`. Comment on your approach and the results in your report.

—— **1-Layer NN** ——
Average Training Loss: 0.5466097997133011
Average Testing Loss: 0.6641005371588785
—— **2-Layer NN** ——
Average Training Loss: 0.48146290601778996
Average Testing Loss: 0.6070283092689537
The TwoLayer achieved lower training than the OneLayer. This is expected: the one-layer network is equivalent to linear regression and can only learn linear mappings, while the two-layer network captures non-linear relationships between features and wine quality. The data has some non-linear structure but is still clearly slightly linear.

For the one layer, a learning rate of 0.001 and 25 epochs was fine likely because the loss surface is convex. The two layer used a learning rate of 0.005, 30 epochs—more passes are needed given the larger parameter space and sigmoid. I think that a model shouldn't be instantly trusted but honestly a human can be wrong.

When the two disagree the context is vital; like who is the human and how much. A company using closed-source software could explain any decision if compelled, whereas a black-box AI model may be genuinely impossible to show even to its creators; making it harder to regulate and track, especially for domains like healthcare or credit where explainability is a legal requirement.